Database Guide

AI Content Creator - backend schema for new developers

PostgreSQL 16 | SQLAlchemy 2.0 models in `backend/app/models/` | schema owned by Alembic (`backend/alembic/versions/`)

13 tables. Every primary key is a UUID v4. Every timestamp is TIMESTAMPTZ (UTC).

1. The big picture

The product is a B2B social-media tool: a team logs in, describes the brands it posts for, writes posts, gets them approved, and publishes them to connected Facebook / Instagram / LinkedIn accounts. The schema follows that story top to bottom:

Layer	Tables	Question it answers
Identity	users, refresh_tokens	Who is logged in?
Tenant	workspaces, workspace_members	Which team, and what may each person do?
Brand profile	brands	For whom is this post written? (tone, audience, logo)
Content	posts, post_media, approval_logs	What is being published, and who signed it off?
Channels	social_accounts, publish_jobs	Where does it go, and did it get there?
Operational	usage_logs, caption_templates, audit_events	Telemetry, seed content, compliance ledger

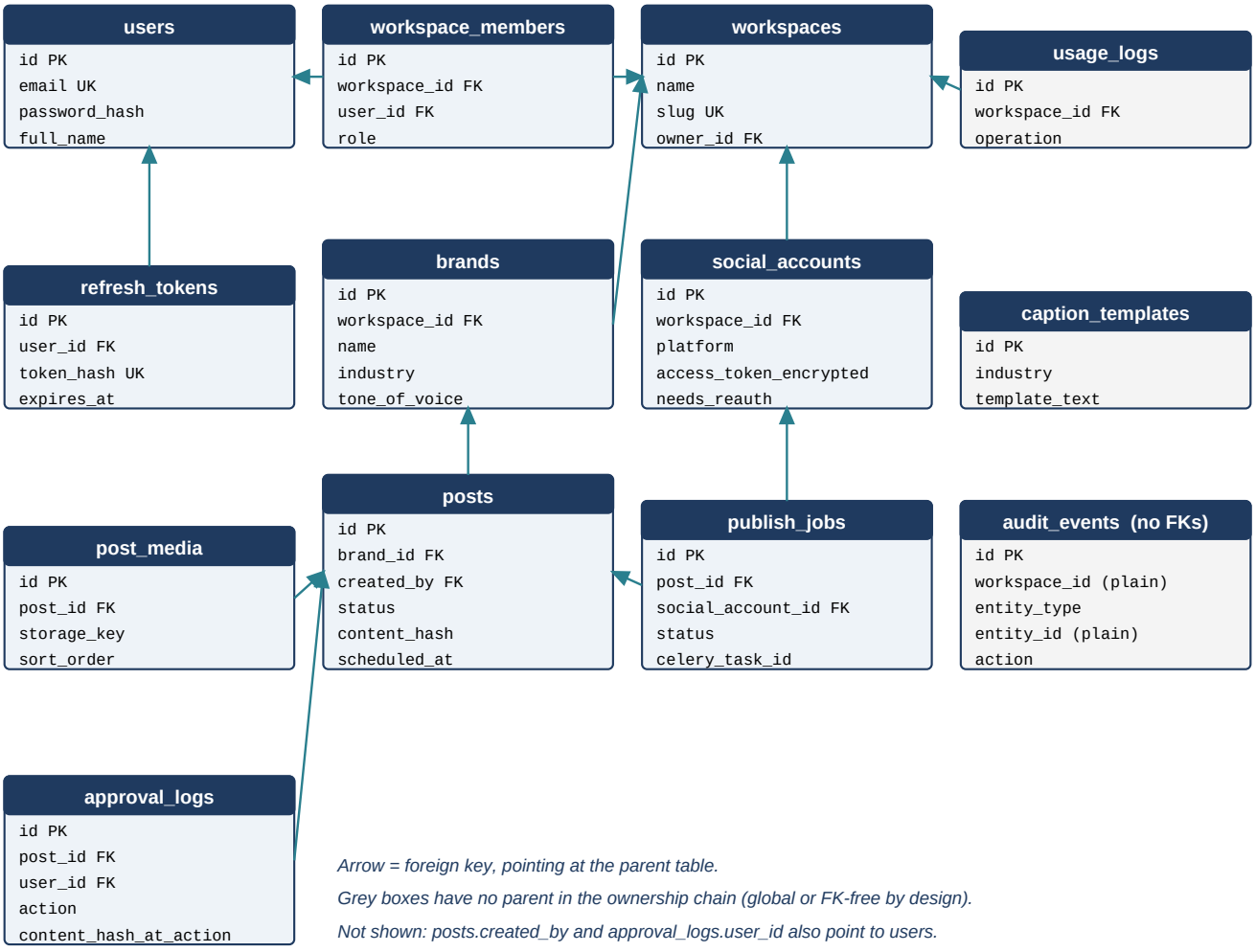
Ownership chain

`workspace -> brand -> post -> post_media / approval_logs / publish_jobs`. Everything a customer owns can be reached from their workspace. Deleting a workspace cascades down this chain (except `audit_events`, which deliberately survives).

Tenancy rule for every query: a user may only touch a row if they are in `workspace_members` for the workspace that row rolls up to. The API enforces this in one place, `app/core/dependencies.py`, never in individual routes.

2. Entity-relationship diagram

Read it top-down: a workspace owns members, brands, channels and telemetry; a brand owns posts; a post owns its media, approval history and publish jobs. publish_jobs is the only table with two parents - it joins a post to the channel it goes out on.



Arrow = foreign key, pointing at the parent table.

Grey boxes have no parent in the ownership chain (global or FK-free by design).

Not shown: posts.created_by and approval_logs.user_id also point to users.

3. Table by table

Each section says what the table is for, what the important columns mean, and how it connects to the rest. Column types are abbreviated; the migration file is the source of truth.

3.1 users

One row per login. Workspaces, posts and approvals all point back here.

Column	Type	Meaning
id	UUID PK	Primary identifier
email	VARCHAR(255) UNIQUE	Login identifier, stored lower-case
password_hash	VARCHAR(255)	bcrypt (cost 12). Never the plain password
full_name, avatar_url	VARCHAR	Display data
is_active	BOOLEAN	Soft disable: an inactive user cannot log in or refresh
created_at, updated_at	TIMESTAMPTZ	Audit timestamps

Relationships: has many refresh_tokens, workspace_memberships; referenced by workspaces.owner_id, posts.created_by, approval_logs.user_id.

3.2 refresh_tokens

Server-side session store for the long-lived login cookie. The raw token lives only in an HttpOnly cookie; the database keeps its SHA-256 hash so a leaked dump cannot replay sessions.

Column	Type	Meaning
user_id	UUID FK -> users (CASCADE)	Token owner
token_hash	VARCHAR(64) UNIQUE	SHA-256 of the raw token
expires_at	TIMESTAMPTZ	Hard expiry (14 days by default)
is_revoked	BOOLEAN	Set on logout and on every rotation (each refresh issues a new row)
ip_address, user_agent	VARCHAR(45), TEXT	For a future 'active sessions' screen

Relationships: belongs to users.

Rotation: /auth/refresh revokes the presented token and inserts a new one. Expect many revoked rows per user; a cleanup job can purge rows where expires_at < now().

3.3 workspaces

The tenant. A workspace is a company or team account: it owns members, brands and connected social accounts. Every user gets one workspace automatically on sign-up.

Column	Type	Meaning
name	VARCHAR(255)	Team or company name
slug	VARCHAR(100) UNIQUE	URL-safe identifier, generated as name + timestamp
owner_id	UUID FK -> users	The founding user; also has an OWNER row in workspace_members

Relationships: has many workspace_members, brands, social_accounts, usage_logs.

3.4 workspace_members

Who belongs to a workspace and with what role. This table is the **authorisation backbone**: every workspace-scoped request is checked against it.

Column	Type	Meaning
workspace_id, user_id	UUID FKs (CASCADE)	UNIQUE together: one role per user per workspace
role	ENUM workspace_role	owner | admin | member
joined_at	TIMESTAMPTZ	When added

Relationships: joins users to workspaces (many-to-many).

Role rules in the API: **owner** can do everything incl. grant admin; **admin** can approve/reject posts, connect accounts, add members; **member** can create brands and posts and submit for approval. An approver can never approve their own post.

3.5 brands

A brand is one business identity the team writes posts for - a client (for an agency) or a product line (for a single company). It carries the style profile that posts, and later the AI, use.

Column	Type	Meaning
workspace_id	UUID FK -> workspaces (CASCADE)	Owning tenant
name, industry	VARCHAR	industry also drives caption_templates suggestions
target_audience	TEXT	Persona summary
tone_of_voice, language	VARCHAR	e.g. 'friendly', 'en'
logo_url	VARCHAR(500)	Brand logo
brand_guidelines	JSONB	Palette, fonts, banned words - free-form
is_active	BOOLEAN	DELETE /brands/{id} deactivates rather than deletes

Relationships: belongs to workspaces; has many posts.

Workspace vs brand: the workspace answers *who is working* (login, roles, connected pages); the brand answers *for whom the post is written*. An agency has one workspace and many brands; a small business usually has one of each.

3.6 posts

The content core. One row per piece of content, carrying its approval state and, after publishing, a filtered summary of the result.

Column	Type	Meaning
brand_id	UUID FK -> brands (CASCADE)	Owning brand (and through it, the workspace)
created_by	UUID FK -> users	Author - used for the 'cannot approve own post' rule
title, caption, hashtags, cta	VARCHAR / TEXT / JSONB	The content. hashtags is a JSON array of strings
status	ENUM post_status	draft | pending_approval | approved | scheduled | publishing | published | partially_published | failed
content_hash	VARCHAR(64)	SHA-256 of caption + sorted media IDs. Approval is bound to this value
version_number	INTEGER	Increments whenever the hash changes
platform_targets	JSONB	Per-platform caption/hashtag overrides (reserved)
scheduled_at, timezone	TIMESTAMPTZ, VARCHAR	When to publish (UTC) and the user's display zone
published_at, publish_result	TIMESTAMPTZ, JSONB	Set by the worker: {platform: {platform_post_id, url}}

Column	Type	Meaning
retry_count	INTEGER	How many worker retries happened

Relationships: belongs to brands; has many post_media, approval_logs, publish_jobs.

Two CHECK constraints guard the state machine at the database level: a *scheduled* post must have scheduled_at, and a publishing/published post must have a content_hash.

3.7 post_media

Images or videos attached to a post. Stores a **storage key**, not a URL - URLs are generated on demand (local path in dev, presigned S3 link in production), so an expiring link never changes the post's content hash.

Column	Type	Meaning
post_id	UUID FK -> posts (CASCADE)	Parent post
storage_key	VARCHAR(1000)	e.g. media/3f9c...a1.png
media_type	ENUM media_type	image | video | carousel_item
sort_order	INTEGER	Carousel order
alt_text	VARCHAR(500)	Accessibility text
metadata	JSONB	Original filename, size, MIME

Relationships: belongs to posts. Adding or removing a row changes the post's content hash and invalidates approval.

3.8 approval_logs

Append-only record of every submit / approve / reject / invalidate on a post, with the content hash at that moment. This is how the system proves *what* was approved and *by whom*.

Column	Type	Meaning
post_id	UUID FK -> posts (CASCADE)	Target post
user_id	UUID FK -> users	Acting user
action	ENUM approval_action	submitted | approved | rejected | invalidated
content_hash_at_action	VARCHAR(64)	Hash of the content when the action happened
rejection_reason	TEXT	Required on reject; also used for the invalidation message

Relationships: belongs to posts.

Never UPDATE or DELETE rows here. Before publishing, the worker recomputes the post's hash and compares it to the latest *approved* row - if they differ, the job fails instead of posting un-reviewed content.

3.9 social_accounts

A connected channel: a Facebook page, Instagram business account or LinkedIn profile. Belongs to the workspace, not to a brand - any brand in the workspace may publish through it.

Column	Type	Meaning
workspace_id	UUID FK -> workspaces (CASCADE)	Owning tenant
platform	ENUM social_platform	facebook | instagram | linkedin
platform_account_id	VARCHAR(255)	The page / profile ID on the platform. UNIQUE with workspace + platform
account_name, account_type	VARCHAR	Display name; page / professional / company
access_token_encrypted	TEXT	OAuth token, Fernet-encrypted at rest (AES-128-CBC + HMAC)

Column	Type	Meaning
refresh_token_encrypted	TEXT	Where the platform supports refresh (LinkedIn, Phase 2)
token_expires_at	TIMESTAMPTZ	Checked every 6 h by a Celery task
platform_metadata	JSONB	page_id, ig_user_id, author_urn - what the adapter needs
is_active	BOOLEAN	Disconnect = set false (row kept so old jobs still resolve)
needs_reauth	BOOLEAN	Set when the token is about to expire; the UI shows a reconnect banner

Relationships: belongs to workspaces; has many publish_jobs.

Tokens are never returned by the API. Decryption happens only inside the publish worker.

3.10 publish_jobs

One row per (post, social account) to publish. Jobs own their own status; the post's status is derived from all of its jobs. A Celery Beat task scans this table every minute and dispatches due rows.

Column	Type	Meaning
post_id	UUID FK -> posts (CASCADE)	What to publish
social_account_id	UUID FK -> social_accounts	Where to publish
celery_task_id	VARCHAR(255)	NULL until dispatched - the scan only picks NULL rows, so nothing runs twice
status	ENUM job_status	queued | processing | completed | failed | cancelled | needs_review
result	JSONB	Filtered platform response: platform_post_id, url, published_at
error_message	TEXT	Why it failed or was cancelled
attempt_number	INTEGER	1 on first run; retries increment
scheduled_for	TIMESTAMPTZ	When it becomes due
started_at, processed_at	TIMESTAMPTZ	started_at is set before the platform call; a job stuck in processing for 1 h is flagged needs_review

Relationships: belongs to posts and social_accounts.

needs_review means 'a worker crashed around the platform call and we do not know if it posted'. A human checks the platform before retrying - this avoids double-posting to a customer's page.

3.11 usage_logs

Lightweight telemetry per workspace: posts created, media uploaded, posts published. Intended for quotas and future billing.

Column	Type	Meaning
workspace_id	UUID FK -> workspaces (CASCADE)	Tenant
user_id	UUID FK -> users, nullable	NULL for system/worker events
operation	ENUM operation_type	post_created | post_published | media_uploaded | ai_generation
metadata	JSONB	post_id, byte counts, etc.

Relationships: belongs to workspaces.

3.12 caption_templates

Global, read-only library of 'Smart Templates' (the MVP stand-in for AI writing). Shared by every tenant, so there is no workspace_id and no write endpoint - content comes from `seeds/seed.py`.

Column	Type	Meaning
industry	VARCHAR(100)	Matched against brands.industry
category	ENUM template_category	promotional | educational | engagement | seasonal
title, template_text	VARCHAR, TEXT	Body with {{placeholder}} tags
suggested_hashtags, suggested_cta	JSONB, VARCHAR	Presets to copy into a post
is_active	BOOLEAN	Hide without deleting

Relationships: no foreign keys - a standalone lookup table.

3.13 audit_events

Universal, tamper-evident ledger of who did what, across every entity. It has **zero foreign keys on purpose**: FK-linked rows are cascade-deleted with their parent, and this table must survive deleting a workspace, user or post.

Column	Type	Meaning
workspace_id, user_id	UUID (no FK)	Plain values - snapshots
user_email, user_name	VARCHAR(255)	Actor snapshot at action time (survives user rename/deletion)
entity_type, entity_id	VARCHAR(50), UUID	Post | Brand | Workspace | SocialAccount | WorkspaceMember
action	ENUM audit_action	CREATE UPDATE DELETE PUBLISH SUBMIT APPROVE REJECT SCHEDULE CONNECT DISCONNECT INVITE ROLE_CHANGE
changes	JSONB	{"before": {...}, "after": {...}}
ip_address, user_agent	VARCHAR(45), TEXT	Forensics

Relationships: none - deliberately. Write via app/services/audit.py::record_audit_event, which never raises.

approval_logs vs audit_events: approval_logs is post-specific and dies with the post; audit_events covers every entity and is permanent. Both are append-only.

4. All relationships in one list

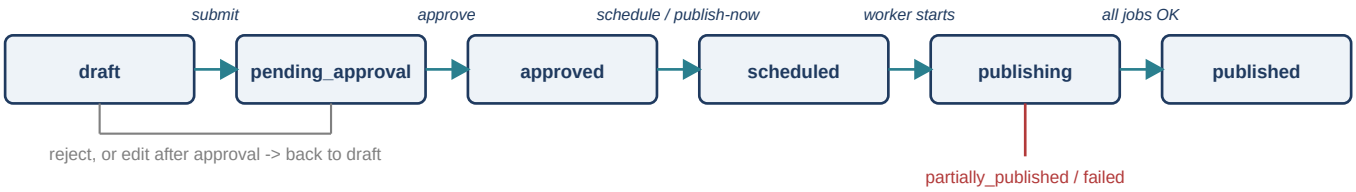
Child.column	Parent	On parent delete	Meaning
refresh_tokens.user_id	users.id	CASCADE	A user's sessions
workspaces.owner_id	users.id	RESTRICT	Founding user (cannot delete a user who owns a workspace)
workspace_members.workspace_id	workspaces.id	CASCADE	Membership
workspace_members.user_id	users.id	CASCADE	Membership
brands.workspace_id	workspaces.id	CASCADE	Brand belongs to a tenant
social_accounts.workspace_id	workspaces.id	CASCADE	Channel belongs to a tenant
posts.brand_id	brands.id	CASCADE	Post belongs to a brand
posts.created_by	users.id	RESTRICT	Author
post_media.post_id	posts.id	CASCADE	Attachment
approval_logs.post_id	posts.id	CASCADE	Approval history
approval_logs.user_id	users.id	RESTRICT	Approver / submitter
publish_jobs.post_id	posts.id	CASCADE	Dispatch unit
publish_jobs.social_account_id	social_accounts.id	RESTRICT	Target channel
usage_logs.workspace_id	workspaces.id	CASCADE	Telemetry
usage_logs.user_id	users.id	RESTRICT	Actor (nullable)
audit_events.*	-	-	No FKs by design
caption_templates	-	-	No FKs; global

RESTRICT here means the database default: the delete fails while children exist. In practice users are deactivated (is_active = false), never deleted.

Unique constraints

Table	Columns	Why
users	email	One account per email
workspaces	slug	URL identifier
refresh_tokens	token_hash	Token lookup
workspace_members	workspace_id + user_id	One role per person per workspace
social_accounts	workspace_id + platform + platform_account_id	Same page connected twice would publish every post twice

5. Post lifecycle and which tables change



Step	API	Rows written
Create	POST /brands/{id}/posts	posts (draft), usage_logs, audit_events
Upload media	POST /media/upload	post_media; posts.content_hash + version_number
Submit	POST /posts/{id}/submit	posts.status=pending_approval, approval_logs(submitted), audit_events
Approve / reject	POST /posts/{id}/approve | reject	posts.status, approval_logs(approved | rejected), audit_events. Approver must be admin/owner and not the author
Schedule / publish now	POST /posts/{id}/schedule | publish-now	posts.status=scheduled + scheduled_at; one publish_jobs row per target account; audit_events
Edit after approval	PATCH /posts/{id}	posts back to draft, approval_logs(invalidated), open publish_jobs -> cancelled
Worker runs	Celery (every minute)	publish_jobs.status processing -> completed/failed; posts.status + publish_result; usage_logs; audit_events

6. Conventions and gotchas

Topic	Rule
Primary keys	UUID v4 generated in Python (uuid.uuid4). Never sequential integers.
Enums	PostgreSQL enum types storing the lower-case VALUE ('draft'), not the Python name ('DRAFT'). Add a value = new Alembic migration with ALTER TYPE ... ADD VALUE.
Timestamps	Always TIMESTAMPTZ, always UTC. posts.timezone is display-only.
JSONB defaults	Server default '{}' or '[]'; never NULL for hashtags, brand_guidelines, metadata.
Soft deletes	brands, social_accounts and users use is_active. posts are hard-deleted (cascade).
Append-only	approval_logs and audit_events: INSERT only. No code path updates or deletes them.
Schema changes	Edit the model in app/models/, then `alembic revision --autogenerate`, review, `alembic upgrade head`. create_all runs only in ENVIRONMENT=development as a convenience.
Two sessions	FastAPI uses the async engine (asyncpg); Celery workers use the sync engine (psycopg2). Write queries in 2.0 style - select(Model).where(...) - so they run on both.
Secrets	Never stored in plain text: passwords are bcrypt hashes, refresh tokens are SHA-256 hashes, OAuth tokens are Fernet-encrypted with FERNET_KEY.

Files to read next

Path	What you will find
backend/app/models/*.py	SQLAlchemy models - the authoritative column list
backend/alembic/versions/0001_initial_schema.py	The migration that creates everything
backend/app/core/dependencies.py	Tenancy checks every route goes through
backend/app/api/v1/posts.py	The approval and scheduling state machine
backend/app/workers/tasks/publish.py	How publish_jobs are dispatched and finalized
backend/seeds/seed.py	Caption templates seed data